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EDUCATION

Technical University of Munich <i>M.Sc. Information Technologies for the Built Environment</i> - Ministry of Education Government Scholarship to Study Abroad (GSSA) - Thesis: Quantifying the Cost of Replacing Registry Metadata with Street-View-Derived Attributes in Archetype-Based Building Energy Modelling	Munich, Germany Oct 2024 - Present
Chung Yuan Christian University <i>B.Arch. Architecture (GPA: 3.77 / 4.0)</i> Graduation Award for Excellent Performance in Professional Studies	Taoyuan, Taiwan Sep 2018 - Jun 2023

PROFESSIONAL EXPERIENCE

SUNHOU Architects & Partners Association <i>Research & Development (Remote)</i> - Leading digital transformation by developing an internal cloud-based RAG system and customizing an ERP system - Developed financial margin dashboards to track labor cost allocation and project progress	Taoyuan, Taiwan Apr 2025 - Present
<i>Project Executive (Working Student)</i> Coordinated public-works projects between office and site teams, managing site progress documentation	Sep 2020 - Jun 2022
Chair of Computing in Civil and Building Engineering, TUM <i>Research Assistant</i> Built a Revit C#/.NET add-in syncing an open BIM model to a version-controlled Neo4j property graph of IFC entities	Munich, Germany May 2026 - Present
ILLOCA <i>BIM Data Specialist (Remote)</i> Constructed ground-truth BIM datasets in Revit for training generative AI models, standardising topological relationships	San Francisco, CA Nov 2025 - Mar 2026

RESEARCH EXPERIENCE

Context-Aware Digital Twin Driven by Hybrid RAG Architected a multi-agent RAG system for digital twins (Python, Neo4j, PostGIS, LangChain) with a full-stack 3D visualisation platform (FastAPI, Next.js, CesiumJS) for context-aware querying	2025
Real-Time Building Digital Model Reconstruction Using a Robotic Agent Deployed ROS 2 SLAM on a Unitree Go2 robot and engineered a Scan-to-BIM pipeline (Point Transformer V3, RANSAC, DBSCAN) producing IFC-compliant models	2025
Agentic Framework for Automated Construction Tendering <i>(AEC Hackathon Munich Edition 2025)</i> Developed an LLM-driven agent with RAG architecture to parse unstructured BoQ PDFs, utilizing DIN 276 standards for semantic trade classification and autonomous vendor matching	2025
Hierarchical Tree Species Classification via Satellite Time-Series Benchmarked Transformers, CNNs, and RNNs against XGBoost and Random Forest on Sentinel-2 multi-spectral time-series, designing hierarchical label structures (leaf type, genus, species) to improve performance and interpretability	2025
Indoor Ambient Monitoring Engineered an IoT-based monitoring system integrating multi-modal sensors (CO, PM2.5, Temperature) via LoRaWAN and OGC SensorThings API for real-time indoor analytics	2025

SKILLS & PROFICIENCY

Domain: ERP systems for architecture firms, project cost & margin control (EVM), BIM (Revit, IFC)
Technical: Python, TypeScript, C#, SQL · PostgreSQL, Neo4j · FastAPI, React, Google Cloud, Docker, Git
Languages: English (Proficient), Mandarin (Native)